TEMASEK
JUNIOR COLLEGE

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Preliminary Examinations 2013

BIOLOGY **9648/03**
Higher 2 **Friday, 19 September**
Paper 3 Applications Paper and Planning Question **2 hours**

Candidate Name: ________________________________ **Civics Group:** __________

READ THESE INSTRUCTIONS FIRST

Write your name and civics group on all answer papers used.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten the answer papers securely and hand up separately.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.
Hand in answer scripts separately.

For Examiner's Use	
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Total	

This document consists of **15** printed pages and 1 blank page.

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Answer **all** questions.

For examiner's use

1 Genetically engineered plants are generated in a laboratory by altering their genetic makeup and are tested in the laboratory for desired qualities. This is usually done by adding one or more genes to a plant's genome using genetic engineering techniques.

Most genetically modified plants are generated by the biolistic method (gene gun) or by *Agrobacterium tumefaciens* mediated transformation. The latter method utilizes the T_i (tumour-inducing) plasmid found in the bacterium, which stimulates the host cells to multiply rapidly, forming tumours. The plasmid contains a short segment called the T-DNA, which is responsible for tumour formation upon integration into the plant genome. (Fig. 1.1)

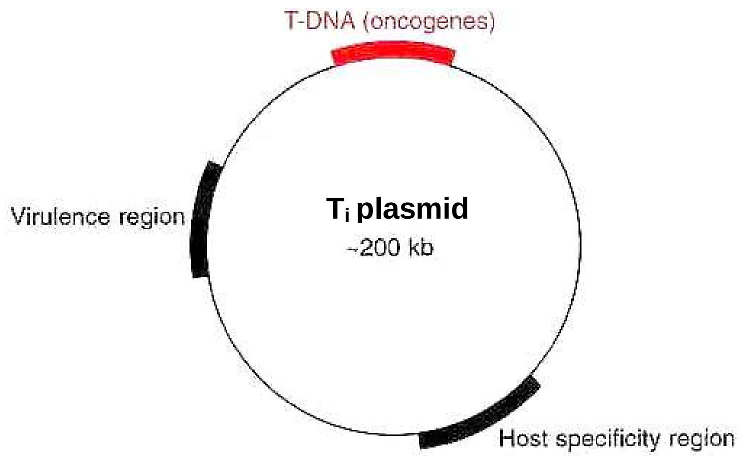

Fig. 1.1

(a) With reference to Fig. 1.1, explain where the gene of interest should be inserted.

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Kan^R gene is usually introduced into the T_i plasmid together with the gene of interest.

(b) Explain the significance of this step.

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For examiner's use

(c) Describe the procedure to insert the gene of interest and the *kan*^{R} gene into the T_i plasmid.

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The New York Times stated that in the early 1990s, Hawaii's papaya industry was facing disaster because of the deadly papaya ringspot virus. Its single-handed savior was a breed engineered to be resistant to the virus. Without it, the state's papaya industry would have collapsed.

Today, 80% of Hawaiian papaya is genetically engineered, and there is still no conventional or organic method to control ringspot virus. Once satisfactory plants are produced, sufficient seeds are gathered, and the companies producing the seed need to apply for regulatory approval to field-test the seeds. If these field tests are successful, the company must then seek regulatory approval for the crop to be marketed.

(d) Explain how plant cloning methods can be beneficial to the production of genetically engineered papaya.

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(e) Suggest why there has been no conventional or organic method that can control the ringspot virus.

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[Total: 12]

For examiner's use

2 Cystic fibrosis is an inherited chronic disease that affects about 70,000 people worldwide. Today, with advances in research and medical technology, patients can expect to live up to 40 years old and beyond. Cystic fibrosis is caused by a single gene mutation.

(a) Describe how gene mutation can result in the development of cystic fibrosis.

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(b) Describe **one** similarity and **one** difference between the treatment of cystic fibrosis and severe combined immunodeficiency using gene therapy.

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In a clinical trial involving 16 patients suffering from cystic fibrosis (CF), liposome-mediated transfer of the normal CFTR gene was carried out using aerosol spray through the nose. Eight patients received the liposome containing normal CFTR gene, while the other 8 patients were given the same treatment without the normal CFTR gene (liposome alone). This latter group is designated as the placebo group.

One of the ways to study the effectiveness of the treatment was to measure the *in vivo* potential difference across the epithelial cell membrane after a low chloride concentration solution was passed through the airway of the lung. Results are shown in Fig. 2.1. Potential difference across a membrane is the measure of concentration of cations and anions inside and outside the cell.

The potential difference is measured 2 days pre-treatment and 2 days post-treatment for each patient and the mean is calculated. The mean potential difference seen in people without CF is also measured.

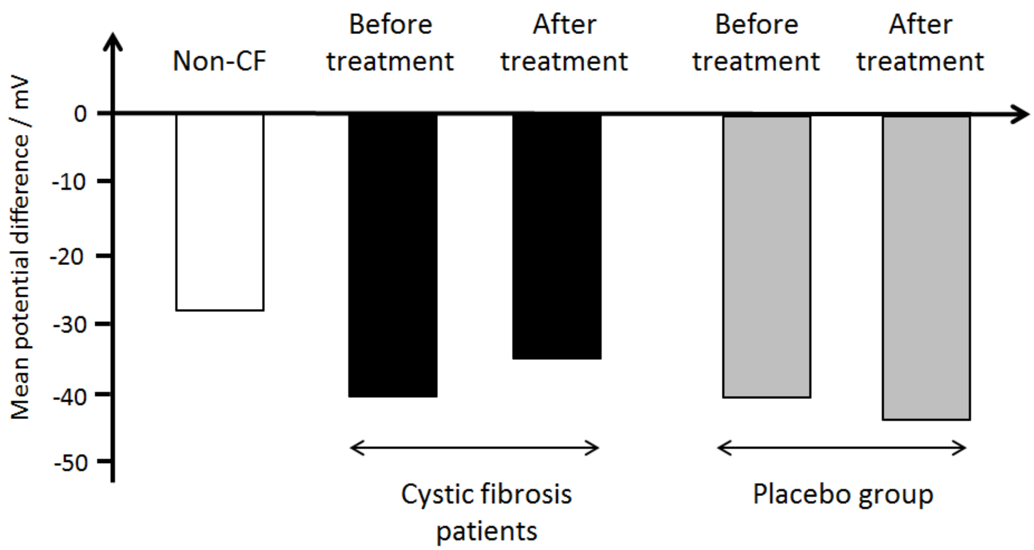

Fig. 2.1

(c) With reference to Fig. 2.1, explain the evidence that shows gene therapy treatment for CF is successful.

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For examiner's use

In liposome-mediated gene therapy of CF, the effectiveness of the treatment is very low, even though an immune response is not generated in patients. Many clinical trials, some of which are described in Table 2.1, have reported low effectiveness even though measurements were taken shortly after treatment.

Table 2.1

Clinical trial / year	**Liposome type**	**No. of patients**	**Target**	**Efficacy**
Caplen et al. / 1995	DC-Chol / DOPE	15	Nose	Potential difference 20% correction towards normal
Zabner et al. / 1997	GL-67™ / DOPE	12	Nose	Partial correction nasal potential difference
Porteous et al. / 1997	DOTAP	16	Nose	Potential difference 20% correction in 2 patients
Noone et al. / 2000	EDMPC-Chol GL-	11	Nose	None
Alton et al. / 1999	DOPE / DMPE-PEG_{5000}	16	Lungs	Potential difference 25% correction towards normal
Ruiz et al. / 2001	DOPE / DMPE-PEG_{5000}	8	Lungs	Vector-specific mRNA in 3 out of 8 patients, no functional protein detected

(d) Suggest **two** reasons why effectiveness of treatment is low when cationic liposomes are used for CF gene therapy.

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The use of gene therapy to treat diseases has raised many ethical and social issues.

(e) Describe **two** issues that have arisen because of gene therapy.

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[Total: 14]

For examiner's use

3 **(a)** Explain the basis of RFLP analysis.

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Three gene loci, **P**, **Q** and **R** are found to be linked, but the order of the genes is not known. Alleles **P**, **Q**, and **R** are found on one chromosome, and alleles **p**, **q** and **r** are found on the other chromosome.

Out of 1000 gametes from an organism, the following genotype frequencies were observed:

Table 3.1

Gamete genotype	Number of gametes
PQR	382
PQr	8
PqR	2
Pqr	98
pQR	105
pQr	1
pqR	9
pqr	395

The recombination frequency between **P** and **R** is 22%.

(b) Determine the recombination frequency between gene loci:

(i) P and Q.

[1]

(ii) Q and R.

[1]

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(c) Indicate on Fig. 3.1 the relative positions of the gene loci **P**, **Q** and **R**.

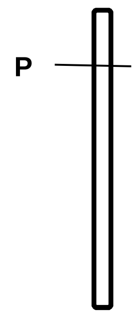

Fig. 3.1

[1]

Restriction mapping is the determination of the positions of restriction sites in a DNA molecule by analyzing the sizes of restriction fragments. Fig. 3.2 shows a gel electrogram after DNA digestion of a 10 kb plasmid, *p*10, using various arrays of restriction enzymes.

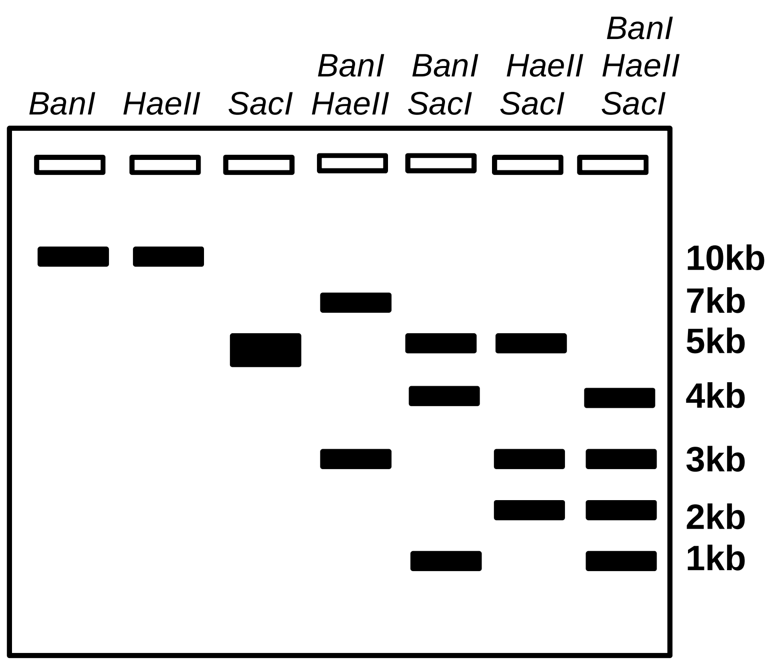

Fig. 3.2
(Adapted from Cornell University http://BioG-1101-1104.bio.cornell.edu)

(d) State **two** other applications of RFLP analysis.

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For examiner's use

Fig. 3.3 shows the plasmid *p*10 with the positions of restriction sites for *Sac*I.

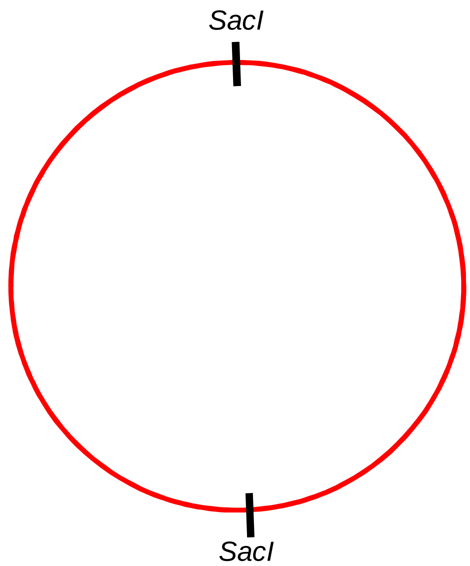

Fig. 3.3

(e) Based on Fig. 3.2, draw the positions of the restriction sites for the **two** other enzymes on Fig. 3.3, and label them accordingly. Include the sizes of the fragments between all the restriction sites. [2]

Smaller DNA molecules tend to have lesser number of restriction sites compared to larger DNA molecules.

(f) Suggest and explain why restriction mapping is less effective with larger DNA molecules.

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[Total: 12]

4 Planning Question

Phospholipases hydrolyse phospholipid bilayer membranes. Once the membrane is hydrolysed, the content within the membrane-bound structure leaks into the surrounding.

To date, many different phospholipases have been isolated, which are specific in the hydrolysis of different phospholipid bilayer membranes, for example, phospholipase A hydrolyses bacterial cell membrane.

In beetroot plants, the large central vacuole contains a water soluble red pigment, betacyanin that gives the beet its characteristic colour. When the integrity of the tonoplast (membrane surrounding the vacuole) is disrupted, the contents of the vacuole will leak into the surrounding environment and this result in a red solution. The composition of the cell membrane and tonoplast is the same. The amount of cellular damage is proportional to the intensity of colour in the solution.

A higher concentration of red solution absorbs more light than one with lower concentration. The absorbance of light can be measured by a spectrophotometer machine set at wave length of 540nm. In a spectrophotometer, light from a light source will pass through a cuvette containing the solution and the light transmitted is received by a detector. The amount of light received is converted into amount of light absorbed by the solution. This is presented as an absorbance value, which has no units.

Using the information and your own knowledge, design an experiment to determine the most effective type of phospholipase for the hydrolysis of tonoplast of beetroot.

You must use:

- Suspension of intact plant cells from beetroot in isotonic solution,
- 1% cellulase solution,
- Different types of phospholipases, each at 1 unit/cm^3 in buffer solution
  - Phospholipase A
  - Phospholipase C1
  - Phospholipase C2
  - Phospholipase D
- 3 cm^3 cuvettes for measuring absorbance using spectrophotometer/colorimeter machine.

You should select from the following apparatus:

- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- syringes,
- pipette fillers,
- white tile or white card,
- timer e.g. stopwatch or stopclock.

Your plan should:

- have a clear and helpful structure such the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

For examiner's use

For examiner's use

For examiner's use

Free Response Question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** In DNA analysis, prolonged incubation of restriction enzymes with plasmid DNA at optimum temperature affects the specificity of the enzyme.

With reference to the principles of gel electrophoresis, discuss why the incubation time is crucial to get accurate results from gel electrophoresis. [9]

(b) Describe the features and functions of zygotic stem cells and embryonic stem cells using **one** named example each. [6]

(c) Using named examples, explain the significance of genetic engineering in solving the demand for food in the world. [5]

[Total: 20]

~ End of Paper 3 ~

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